

The role of thermodynamic forcing in direct numerical simulations of turbulence-cloud interactions

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Particle-based direct numerical simulation (DNS) is a promising approach to study microscale turbulence-cloud-aerosol interactions. Such a model requires external forcing to sustain turbulent fluctuations in the momentum and thermodynamic fields. Although momentum forcing is common in particle-based DNS studies of atmospheric clouds, thermodynamic forcing has not gained enough attention. This study implements external forcing in the thermodynamic fields of temperature and vapor mixing ratio in addition to momentum forcing. The forcing terms are constructed in the Fourier space using a band of low wavenumbers. The turbulent flow of air is assumed to be homogeneous and isotropic. Aerosol particles and cloud droplets are tracked in Lagrangian fashion. The entrainment of ambient air into the cloud and subsequent turbulent mixing are simulated. It is found that thermodynamic forcing sustains fluctuations in the thermodynamic fields and reduces the mean temperature, vapor mixing ratio, and supersaturation. Although thermodynamic forcing is applied at large scales only, it influences fluctuations at all scales by interacting with latent heat release or absorption. The probability distributions of forced thermodynamic fields are negatively skewed with kurtosis values greater than three. With regards to microphysics, thermodynamic forcing suppresses condensation and activation but enhances evaporation and deactivation. It is shown that thermodynamic forcing has a broadening effect on the size distributions of particles during condensation and evaporation.

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NOMENCLATURE

c_p, C_d	= specific heat of air [J/(kg.K)], time rate of exchange of mass [1/s]
δ_{k,k_f}, η	= delta function [-], Kolmogorov length scale [m]
$\epsilon_{T,in}, \epsilon_{q_v,in}$	= dissipation rates in the initial temperature [K ² /s], and vapor [(g/kg) ² /s] fields
ϵ	= turbulent kinetic energy dissipation rate [m ² /s ³]
$\mathbf{F}_b, \mathbf{F}_c$	= buoyancy force [m/s ²], external force [m/s ²] in the momentum field per unit mass
F_T, F_{q_v}	= external forcing in the temperature [K/s], water vapor [(g/kg)/s] fields
$\mathbf{g}, \mathbf{k}_f$	= acceleration vector due to gravity [m/s ²], forcing wavenumber vector [1/m]
k_{min}, k_{max}	= minimum wavenumber [1/m], maximum wavenumber [1/m]
κ, k_T	= hygroscopicity of solute [-], thermal conductivity in air [W/(m.K)]
l_0, l	= length scale characterizing large eddies [m], integral length scale [m]
L_h, λ, M_l	= specific latent heat [J/kg], Taylor microscale [m], molar mass of water [kg/mol]
m_a, μ_T, μ_v	= mass of air [kg], molecular diffusivities [m ² /s] of temperature, water vapor in air
ν, p	= kinematic viscosity of air [m ² /s], instantaneous fluid pressure [N/m ²]
$q_v, q_{v,s}$	= instantaneous water vapor mixing ratio [g/kg], saturation vapor mixing ratio [g/kg]
r_d, r	= dry aerosol radius [μm], wet radius of aerosol particle or cloud droplet [μm]
r_c, R_v	= critical radius [μm], specific gas constant for water vapor [J/(kg.K)]
$R_{F_{q_v}C_d}$	= cross-correlation coefficient between vapor forcing and condensation rate [-]
ρ_a, ρ_l, S_e	= density of air [kg/m ³], density of water [kg/m ³], environmental supersaturation [-]
S_k, S_c	= particle equilibrium supersaturation [-], critical equilibrium supersaturation [-]
S_t, σ_l	= Stokes number [-], surface tension of water [N/m]
T, τ_p	= instantaneous temperature [K], particle response time [s]
$\mathbf{u}, u_{rms}$	= instantaneous fluid velocity vector [m/s], root-mean-square velocity [m/s]
$\mathbf{V}, \mathbf{X}$	= particle velocity vector [m/s], particle position vector [m]
PDF, TKE	= probability density function [-], turbulent kinetic energy [m ² /s ³]

I. INTRODUCTION

Numerical modeling has become a powerful tool to investigate turbulence-cloud-aerosol interactions in the Earth's atmosphere as experiments and measurements are difficult and expensive to conduct. The most accurate approach is direct numerical simulation (DNS) coupled with Lagrangian particle tracking. This method resolves the flow of turbulent air up to the smallest scale without any kind of parameterizations of small turbulent eddies and tracks cloud droplets and aerosol particles individually. Two thermodynamic (scalar) fields, namely temperature and water vapor mixing ratio are transported in the Eulerian framework. Flow turbulence is assumed to be homogeneous and isotropic, but it decays in absence of a turbulence production mechanism. To prevent the decay, many DNS studies (Vaillancourt,¹ Kumar et al.,² Li³) and our previous DNS studies (Gao et al.,⁴ Sharfuddin et al.⁵) applied external forcing in the momentum field only. Although velocity turbulence was sustained, thermodynamic fluctuations decayed in those studies. However, thermodynamic fluctuations are sustained in the atmosphere by mechanisms such as strong entrainment of ambient air into the cloud and solar radiation absorbed by the cloud.^{6,7,8} Numerical representation of these events requires addition of external forcing in the thermodynamic fields.

The most common method of forcing in DNS models is Fourier forcing. The idea is to inject energy in a band of low wavenumbers (large scales) which cascades down to small scales. Eswaran and Pope⁹ showed that Fourier forcing in the momentum field results in a statistically stationary velocity field. Bogucki et al.¹⁰, Watanabe and Gotoh¹¹, and Chen and Cao¹² applied Fourier forcing in a passive scalar (thermodynamic) field besides momentum forcing. They suggested that the scaling behavior of a passive scalar differs from that of velocity because scalar fluctuations are highly intermittent at small scales and become decorrelated more rapidly than velocity fluctuations do. In the context of atmospheric clouds, the DNS model of Paoli and Shariff¹³ implemented Fourier forcing in the momentum, temperature, and water vapor fields. They reported that the temperature and vapor fields reach statistical stationarity due to external forcing. The DNS model of Chen et al.¹⁴ forced the thermodynamic fields in a low wavenumber band to simulate the growth of mixed-phase clouds. Satio et al.¹⁵ applied Fourier forcing in the water vapor field but kept the temperature field unforced. Their DNS study showed that water vapor forcing could maintain fluctuations in the supersaturation at a desired level.

External thermodynamic forcing influences microscale phase change in numerical models. Paoli and Shariff¹³ suggested that the temperature and vapor fluctuations sustained by forcing broaden the size distributions of cloud droplets. Satio et al.¹⁵ found that droplet size distributions are narrower in their DNS model relative to those observed

in the experiments if thermodynamic forcing is absent. The large eddy simulation (LES) model of Endo et al.¹⁶ applied external forcing in the temperature and water vapor fields. The authors reported that the total mass of cloud droplets and their number concentration vary in response to forcing. Fu et al.¹⁷ and Shen et al.¹⁸ used temperature and vapor forcings in their 2D cloud-resolving model which has a larger domain size and coarser grid resolution than typical DNS and LES studies. They found that large-scale forcing could enhance or suppress cloud growth and precipitation depending on the details of simulation parameters. The LES model of Thomas et al.¹⁹ showed that temperature and vapor forcings make the probability distributions of supersaturation negatively skewed and affect supersaturation-microphysics interactions. Using a simplified one-dimensional model, Wu and Moncrieff²⁰ reported that large-scale forcing of the temperature and water vapor is necessary to assess the impact of mesoscale circulation on microscale cloud processes.

Despite the abovementioned progress, knowledge gaps still exist regarding the role of thermodynamic forcing in microscale cloud models. Specifically, the interaction between latent heat release or absorption and thermodynamic forcing needs better understanding. Also, microphysical processes like condensation, aerosol activation, evaporation and cloud droplet deactivation should be simulated with persistent thermodynamic fluctuations. Hence, we extend our previous DNS study⁵ to include external forcing in the temperature and vapor mixing ratio fields. The forcing terms are constructed in the Fourier space,⁹ and forcing is imposed at large scales or low wavenumbers only. External forcing is also present in the momentum field. The mutual conversion of aerosol particles and cloud droplets is modeled using the kappa-Kohler theory. Simulations are conducted in three dimensions for two sets of initial conditions. We vary the magnitude of thermodynamic and momentum forcing terms in our DNS model. The results are compared for the thermodynamics and microphysics of cloud. The scale dependence of thermodynamic fluctuations as well as their probability distributions are examined. The size distributions of particles are calculated to quantify their broadening or narrowing in response to external forcing.

II. GOVERNING EQUATIONS

The equations solved in our model are described in this section. They include the Navier-Stokes equations, the transport equations for thermodynamic variables, the equations for external forcing, Lagrangian particle tracking, and the kappa-Kohler theory.

A. Turbulent velocity field and momentum forcing

The turbulent flow of ambient air is assumed to be governed by the Navier-Stokes equations with the Boussinesq approximation:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ is the instantaneous fluid (air) velocity vector and p is the instantaneous fluid pressure. The $\mathbf{F}_b$ is the buoyancy force and can be expressed as:²¹

$$\mathbf{F}_b = -g \left[\frac{T - \langle T \rangle}{\langle T \rangle} + 0.608(q_v - \langle q_v \rangle) - q_c \right]. \quad (3)$$

where T is the instantaneous temperature, q_v and q_c are the instantaneous water vapor mixing ratio and liquid water mixing ratio,⁵ respectively. $\langle T \rangle$ and $\langle q_v \rangle$ are the domain averages of T and q_v , respectively. Our model's domain size (0.512 m) is much smaller compared to an actual cloud and thus $\mathbf{F}_b$ is found to be negligible. Since $\mathbf{F}_b$ cannot prevent a complete decay of the imposed turbulent velocity field, momentum forcing ($\mathbf{F}_c$) is needed in Eq. (1). The $\mathbf{F}_c$ can be formulated in the Fourier space as:²²

$$\hat{\mathbf{f}}_c(\mathbf{k}, t) = \epsilon_{in} \frac{\hat{\mathbf{u}}(\mathbf{k}, t)}{\sum_{\mathbf{k}_f \in \mathbf{k}} |\hat{\mathbf{u}}(\mathbf{k}_f, t)|^2} \delta_{\mathbf{k}, \mathbf{k}_f}. \quad (4)$$

where ϵ_{in} is an input TKE dissipation rate, $\mathbf{k}$ is the wavenumber vector, $\hat{\mathbf{u}}(\mathbf{k}, t)$ is the Fourier-transformed velocity field, $\mathbf{k}_f$ is the forced wavenumber vector, and $\delta_{\mathbf{k}, \mathbf{k}_f}$ is a delta function which is unity inside the wavenumber shell consisted of the forced wavenumbers but zero otherwise. The $\hat{\mathbf{f}}_c$ is inverse transformed into the physical space as $\mathbf{F}_c$.

B. Transport and forcing of the thermodynamic fields

The transport equations for temperature (T) and water vapor mixing ratio (q_v) are written as:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T + F_T, \quad (5)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_v \nabla^2 q_v + F_{q_v}. \quad (6)$$

where C_d quantifies the time rate of exchange of mass between liquid water and water vapor and is described by Eq. (17). $L_h C_d / c_p$ and $-C_d$ represent the phase change terms in Eqs. (5) and (6), respectively. μ_T and μ_v are the molecular diffusivities of temperature and water vapor, respectively. F_T and F_{q_v} are the external forcing terms in the temperature

and vapor mixing ratio fields, respectively. Let θ be a placeholder for T and q_v and F_θ be the thermodynamic forcing term in a general sense. Then F_θ can be constructed in the Fourier space as:

$$\hat{f}_\theta(\mathbf{k}, t) = \epsilon_{\theta, in} \frac{\hat{\theta}(\mathbf{k}, t)}{\sum_{\mathbf{k}_f \in \mathbf{k}} |\hat{\theta}(\mathbf{k}_f, t)|^2} \delta_{\mathbf{k}, \mathbf{k}_f}. \quad (7)$$

Above, $\hat{\theta}(\mathbf{k}, t)$ is the Fourier-transformed thermodynamic field and $\epsilon_{\theta, in}$ is an input dissipation rate for θ . The $\hat{f}_\theta(\mathbf{k}, t)$ is converted back to the physical space and becomes F_θ . Note that the F_θ follows a Gaussian distribution and the domain-average of F_θ is zero. Also, external forcing terms in the momentum and thermodynamic fields are randomly generated and they are independent of each other.

Growth of cloud droplets and aerosol particles

The motion of each aerosol particle or cloud droplet is represented in the Lagrangian fashion. The simplified expressions for the Lagrangian position and velocity are²

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad (8)$$

$$\frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}. \quad (9)$$

where the subscript 'i' denotes the i-th aerosol particle or cloud droplet, $\mathbf{V}$ is its velocity vector, and $\mathbf{X}$ is its position vector. $\mathbf{u}_i$ is the instantaneous fluid velocity vector at particle position i and is obtained through a trilinear interpolation of the Eulerian field. The particle response time τ_p is given by²

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}. \quad (10)$$

Equation (9) is valid when the particle Stokes number (S_t) is significantly less than unity. The time evolution of particle radius (r) is described by²³

$$\frac{dr_i(t)}{dt} = \frac{G_i}{r_i(t)} (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (11)$$

where S_e is the environmental supersaturation and S_k is the particle equilibrium supersaturation that accounts for the curvature and solute effects combined. The kappa-Kohler theory²⁴ expresses the S_k as:

$$S_k = \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1 - \kappa)}, \quad A = \frac{2\sigma_l M_l}{RT\rho_l}. \quad (12)$$

where r_d is the dry aerosol radius, κ is the hygroscopicity of solute (dry aerosol), R is the universal gas constant, σ_l is the surface tension of water, and M_l is molar mass of water. If the atmosphere is supersaturated, aerosol particles grow

beyond the critical radii and activate into cloud droplets. Whereas cloud droplets shrink below the critical radii and deactivate into aerosol particles in a subsaturated environment. Using the kappa-Kohler theory, the critical radius (r_c) and critical equilibrium supersaturation (S_c) are derived as:²⁵

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}, \quad S_c = \frac{2}{\sqrt{\kappa}} \left(\frac{A}{3r_d} \right)^{\frac{3}{2}}. \quad (13)$$

A particle can be classified as aerosol particle or cloud droplet by comparing its wet radius (r) with the r_c . An aerosol particle corresponds to $r_d < r \leq r_c$ while a cloud droplet corresponds to $r > r_c$. The diagnostic variable S_e depends on the prognostic variables q_v and T and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1, \quad q_{v,s} = 621.97 \frac{p_{sat}}{p - p_{sat}}, \quad (14)$$

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right). \quad (15)$$

$q_{v,s}$ is the saturation vapor mixing ratio²⁶ and p_{sat} is the saturation vapor pressure.²⁷ The growth parameter G is found as:²³

$$G = \frac{1}{\frac{L_h \rho_l}{k_T T} \left(\frac{L_h}{R_v T} - 1 \right) (1 + S_k) + \frac{\rho_l R_v T}{\mu_v p_{sat}}}, \quad (16)$$

where k_T is the thermal conductivity in air and R_v is the specific gas constant for water vapor. The C_d is determined as:

$$C_d(\mathbf{x}, t) = \frac{4\pi\rho_l}{\rho_a a^3} \sum_{i=1}^n G_i r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (17)$$

where n is the number of particles in a grid cell that has a volume of a^3 .

III. NUMERICAL DETAILS

We run simulations for two scenarios: Scenario I and Scenario II.⁵ The first is for condensation and activation of aerosol particles into cloud droplets, while the second is for evaporation and deactivation of cloud droplets into aerosol particles. Since the domain size of our DNS model is much smaller than that of a natural cloud, it will be difficult to observe significant condensation and evaporation at the same time in our model. So, we design two separate scenarios that correspond to different initial conditions. In the first scenario, the cloud volume mixes with saturated ambient air, and condensation prevails. In the second, the dry (subsaturated) air is entrained into the cloud volume and evaporation is dominant.

A. Momentum, thermodynamic, and particle fields

We vary the range of forced wavenumbers to generate two levels of flow turbulence: ‘low’ and ‘high’ and denote them as ‘L’ and ‘H’, respectively.⁵ The forcing wavenumber bands are: $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 16.18$, respectively. The minimum wavenumber $k_{min} = 2\pi/L = 2\pi/0.512 \sim 12.27$ and the maximum wavenumber $k_{max} \sim 128k_{min}$. The initial TKE spectrum is specified as²⁸

$$E(k) = \frac{16}{\sqrt{\pi/2}} \frac{u_0^2 k^4}{k_0^5} \exp\left(-\frac{2k^2}{k_0^2}\right). \quad (18)$$

where $k_0 = 4.12k_{min}$ and $u_0 = 0.16 \text{ m/s}$ is the root-mean-square (rms) velocity. Table I lists the turbulence statistics such as the turbulent kinetic energy dissipation rate, ϵ , the Taylor micro-scale, λ , the Kolmogorov length scale η , the length scale characterizing large eddies, l_0 , the forcing Reynolds number, Re_f , the Taylor-scale Reynolds number, Re_λ , and the Reynolds number based on l_0 , Re_{l_0} . The definition of abovementioned quantities can be found in Ref. 29.

Table I. Statistics of the turbulent velocity field at 4 s

Turbulence level	ϵ (m^2/s^3)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
High	0.00723	0.0169	0.00082	0.184	455.41	156.11	1704
Low	0.00063	0.0261	0.00153	0.227	201.90	138.74	1207

Our model resembles a 3D cube inside which flow turbulence, cloud droplets, and aerosol particles interact. The boundary conditions are triply periodic, and the domain volume is 0.512^3 m^3 . At the start of simulations, the domain is divided equally into cloudy and clear-air regions. A slab-like configuration⁴ is assumed for the initial field (Fig. 1) to enable a sharp transition in the water vapor mixing ratio and temperature at the cloud/clear-air interface. The initial thermodynamic fields change abruptly across the interface in our model unlike those in Paoli and Shariff¹³ and Chen et al.¹⁴ Such an inhomogeneous distribution of the initial thermodynamic fields allows us to realistically simulate turbulent mixing between the cloud and environment.³⁰

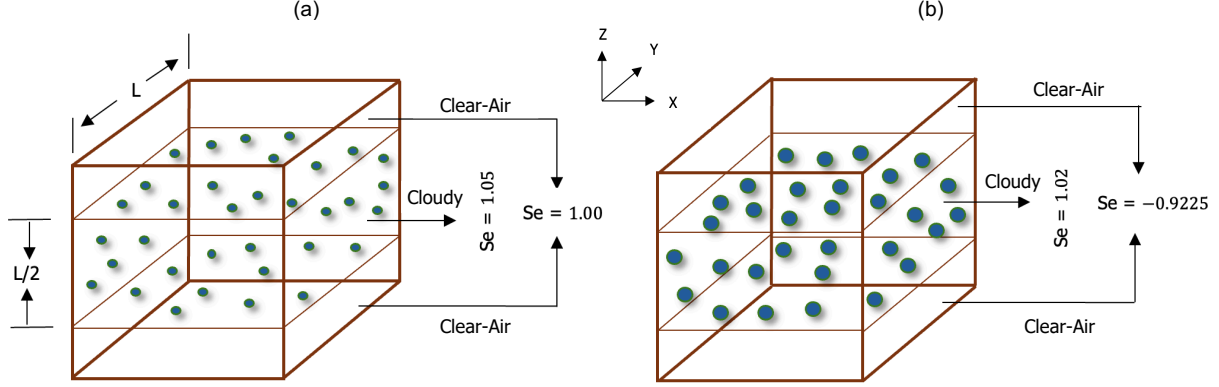


FIG. 1 Initial configurations in (a) Scenario I, and (b) Scenario II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. A sharp interface separates the two regions. The cloudy region is initially 5% supersaturated in Scenario I and 2% in Scenario II. The clear-air region is just saturated in the first and 92.25% subsaturated in the second. The spheres represent dry aerosols in Scenario I and cloud droplets in Scenario II.

In Scenario I, the cloudy region is supersaturated (humid), but the clear-air region is just saturated at initial time.

The initial vapor mixing ratio field is specified as:⁴

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases} \quad (19)$$

Above, $q_v^{max} = 4.066 \text{ g/kg}$, and $q_{v,e} = 3.872 \text{ g/kg}$. The length, $d \equiv L/2$ is the width of the cloud slab. q_v^{max} and $q_{v,e}$ belong to the cloudy and clear-air regions, respectively. The temperature field is initialized as:⁴

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (20)$$

Based on Eqs. (19) and (20), the mean temperature and mean water vapor mixing ratio are not zero unlike the mean velocity. Since the initial q_v and T follow a discontinuous distribution, we choose a finite difference approach⁵ instead of a pseudospectral method as the latter could give rise to artificial oscillations (Gibbs phenomenon).² We simulate five cases in Scenario I which are summarized in Table II. The forcing wavenumber bands corresponding to ‘low’ and ‘high’ levels of thermodynamic forcing are: $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 16.18$, respectively. At the start of simulations, 16×10^6 dry aerosols are randomly placed in the cloudy region, giving a number concentration of 119 cm^{-3} . The size (radius) distribution of dry aerosols is assumed to be monodisperse with a mean of $0.1 \mu\text{m}$. Dry aerosols absorb water from the environment to become aqueous aerosol particles which grow further and activate into cloud droplets in Scenario I.

TABLE II. Test cases in Scenario I and Scenario II.

Case	Dry aerosol size (μm)	TKE dissipation rate (m^2/s^3)	Level of momentum forcing (F_C)	Level of thermodynamic forcing (F_θ)	Initial particle size (μm) in Scenario I	Initial particle size (μm) in Scenario II
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L	0.1	0.00063	Low	N/A	0.1	15
L-L	0.1	0.00063	Low	Low	0.1	15
L-H	0.1	0.00063	Low	High	0.1	15
H	0.1	0.00723	High	N/A	0.1	15
H-L	0.1	0.00723	High	Low	0.1	15
Case	Comments					
L	Level of momentum forcing is low (L). Thermodynamic forcing is absent or $F_T = F_{qv} = 0$.					
L-L	Level of momentum forcing is low (L). Level of thermodynamic forcing is low (L).					
L-H	Level of momentum forcing is low (L). Level of thermodynamic forcing is high (H).					
H	Level of momentum forcing is high (H). Thermodynamic forcing is absent or $F_T = F_{qv} = 0$.					
H-L	Level of momentum forcing is high (H). Level of thermodynamic forcing is low (L).					

In Scenario II, the cloudy region is supersaturated (humid) but the clear-air region is subsaturated (dry) at initial time with $q_v^{max} = 3.949 \text{ g/kg}$ and $q_{v,e} = 0.30 \text{ g/kg}$. A total of 16×10^6 cloud droplets, each of them having a radius (r) of $15 \text{ }\mu\text{m}$, are randomly placed in the cloudy region at the start. Dry aerosols, also known as solutes, are assumed to be present in the core of cloud droplets with $r_d = 0.1 \text{ }\mu\text{m}$. For a known r_d , the r_c can be calculated using Eq. (13). The r_d and r_c are fixed at each particle position while the wet radii (r) evolve with time. We again simulate five cases (Table II) in Scenario II which correspond to evaporation and deactivation of cloud droplets.

B. Parameters, grid resolution, and computing resources

Table III lists the constants and parameters used in our model including the domain-averaged temperature, pressure, and Stokes number (S_t) at initial time. The chemical composition of dry aerosol is ammonium sulfate with a hygroscopicity (κ) value of 0.61.³¹ Particles are allowed to grow (Eq. (11)) after the TKE dissipation rate becomes statistically stationary which takes 2 s. The input dissipation rates in the temperature ($\epsilon_{T,in}$) and vapor mixing ratio ($\epsilon_{qv,in}$) fields and the Stokes numbers at initial time are shown in Table III. Note that the mean temperature, pressure, and density do not vary with height in our microscale model. A detailed analysis in this regard can be found in Ref. 5.

TABLE III. Parameters and constants

Symbol	Value	Unit	Symbol	Value	Unit
q_v^{max}	4.066 (Scenario I)	g/kg	$q_{v,e}$	3.872 (Scenario I)	g/kg
	3.949 (Scenario II)			0.30 (Scenario II)	
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	0.92	kg/m ³	ν	1.5×10^{-5}	m ² /s
ρ_l	1000	kg/m ³	$\langle p(t = 0) \rangle$	71446	N/m ²

$\langle T(t=0) \rangle$	270.75	K	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	σ_l	0.072	N/m
$\mu_T = \mu_v$	2.2×10^{-5}	m ² /s	M_l	0.018	kg/mol
R	8.314	J/mol/K	$\langle S_t(t=0) \rangle$	9.6×10^{-7} (Scenario I)	-
κ	0.61	-		2.16×10^{-2} (Scenario II)	
$\epsilon_{T,in}$	1×10^{-5} (Scenario I)	K ² /s	$\epsilon_{qv,in}$	1×10^{-4} (Scenario I)	(g/kg) ² /s
	1×10^{-3} (Scenario II)			1×10^{-2} (Scenario II)	

To balance the required grid resolution and computational cost, we use 256^3 grid points for all simulations. The $k_{max}\eta$ values for ‘low’ and ‘high’ turbulence levels are 2.46 and 1.43, respectively. The details about the numerical solvers and software package are available in Ref. 5. The Perlmutter supercomputer, managed by the National Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA, has been used for the simulation purposes.

IV. RESULTS AND DISCUSSIONS

We discuss the impact of external forcing on the fluctuating and mean thermodynamic fields in Sec. IV A. The impact of thermodynamic forcing on condensation, aerosol activation, evaporation, and cloud droplet deactivation is explained in Sec. IV B. The concluding remarks in Sec. V summarize our findings.

A. The impact of external forcing on thermodynamic fields

The temporal evolution of the variances of temperature (σ_T^2) and vapor mixing ratio (σ_{qv}^2) is shown in Figs. 2(a)-(d). We see that thermodynamic fluctuations attain statistical stationarity when forcing is applied (Cases L-L, L-H, and H-L) but decay otherwise (Cases L and H). The thermodynamic fields are inhomogeneous at initial time, marked by a sudden change at the interface. But they become more homogeneous with time, and their variances diminish in absence of thermodynamic forcing (Cases L and H). A higher level of momentum forcing enhances mixing and phase change, resulting in a faster decay of variance for Case H compared to Case L (Figs. 2(a)-(d)). On the other hand, thermodynamic forcing acts as a source of fluctuations and counteracts the decay (Cases L-L, L-H, and H-L). Flow turbulence (momentum forcing) does not dissipate or augment thermodynamic fluctuations but propagates them from one place to another in our model while thermodynamic forcing generates fluctuations locally. The magnitude of temperature fluctuations is larger in Scenario II (Fig. 2(c)) compared to Scenario I (Fig. 2(a)) and the same is true for water vapor fluctuations (Fig. 2(d) vs Fig. 2(b)).

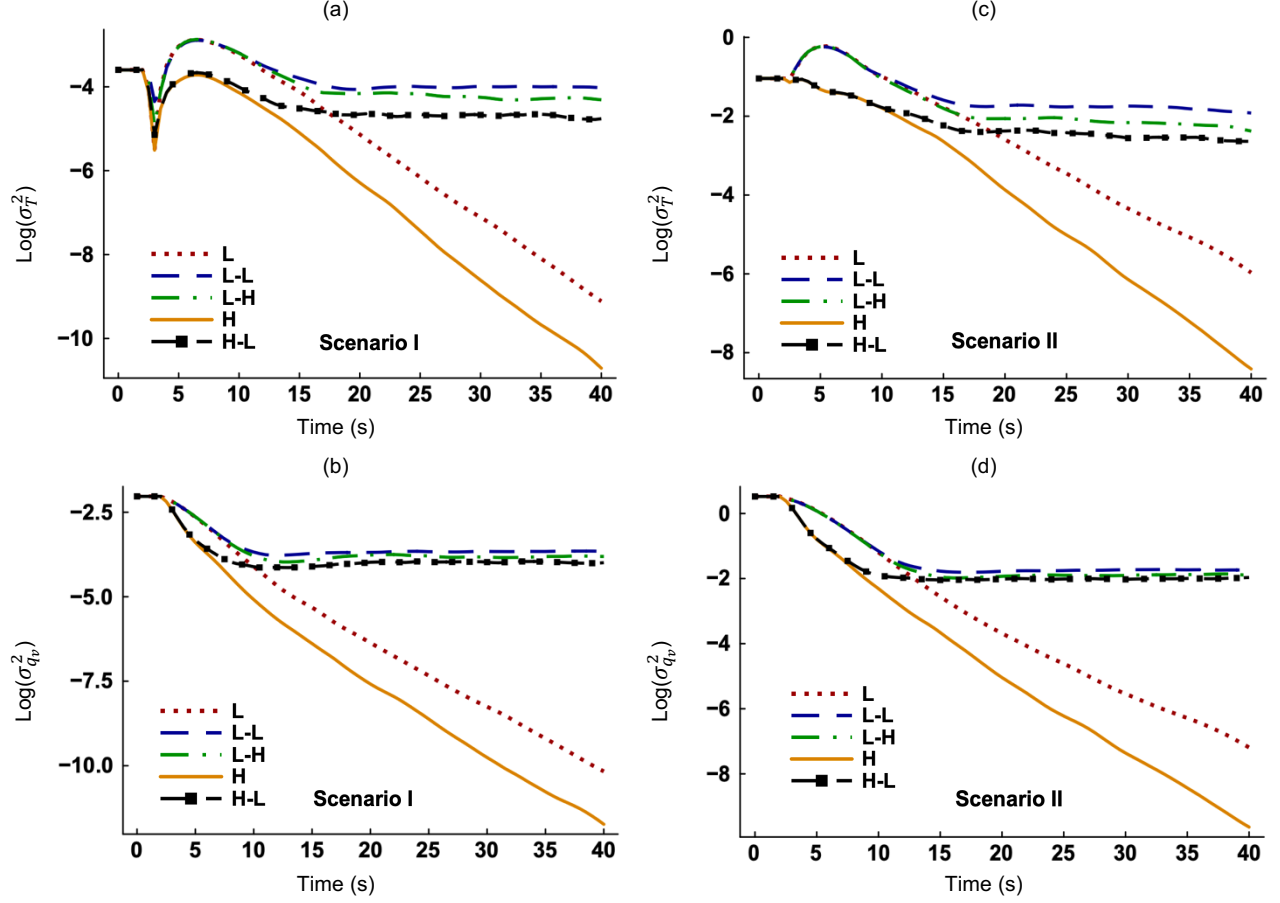


FIG. 2 Time evolution of the variances of (a) temperature and (b) water vapor mixing in Scenario I, respectively. Time evolution of the variances of (c) temperature and (d) water vapor mixing in Scenario II, respectively.

The energy spectra of temperature ($E_T(k)$) and vapor mixing ratio ($E_{q_v}(k)$) are shown in Figs. 3(a)-(b) for the five cases at 25 s in Scenario I. Only momentum forcing is present for Cases L and H, and we see that the $E_T(k)$ and $E_{q_v}(k)$ differ at large scales but converge at small scales. This is consistent with turbulence theory as kinetic energy injected at large scales breaks down into progressively smaller scales, leading to universal statistics at small scales.³² The areas under the curve (Figs. 3(a)-(b)) appear to be larger for Case L relative to Case H because the magnitude of fluctuations is larger for the former at 25 s (Fig. 2). Fluctuations increase at all scales for thermodynamically forced cases (L-L, L-H, and H-L) compared to the unforced cases (L and H). Since thermodynamic forcing is applied at large scales only, the lack of convergence at small scales is a deviation from Kolmogorov's hypothesis. This happens due to the interaction between thermodynamic forcing (F_θ) and phase change (C_d) terms in Eqs. (5) and (6). To understand the role of C_d in thermodynamic spectra, we simulate two more cases: LA and LA-L by setting $C_d = 0$. Thus, Case LA is Case L without phase change and Case LA-L is Case L-L without phase change. We report in Fig. 3(c) that fluctuations are significantly reduced at large and intermediate scales for Case L compared to Case LA although a

marginal increase is visible at small scales. Hence, inclusion of phase change reduces thermodynamic fluctuations in our model as latent heat release or absorption dampens thermodynamic gradients. On the contrary, thermodynamic forcing increases fluctuations at large scales only if the C_d is absent (Cases LA-L vs LA). Based on Fig. 3(c), the C_d and F_θ have contrasting effects on thermodynamic fluctuations.

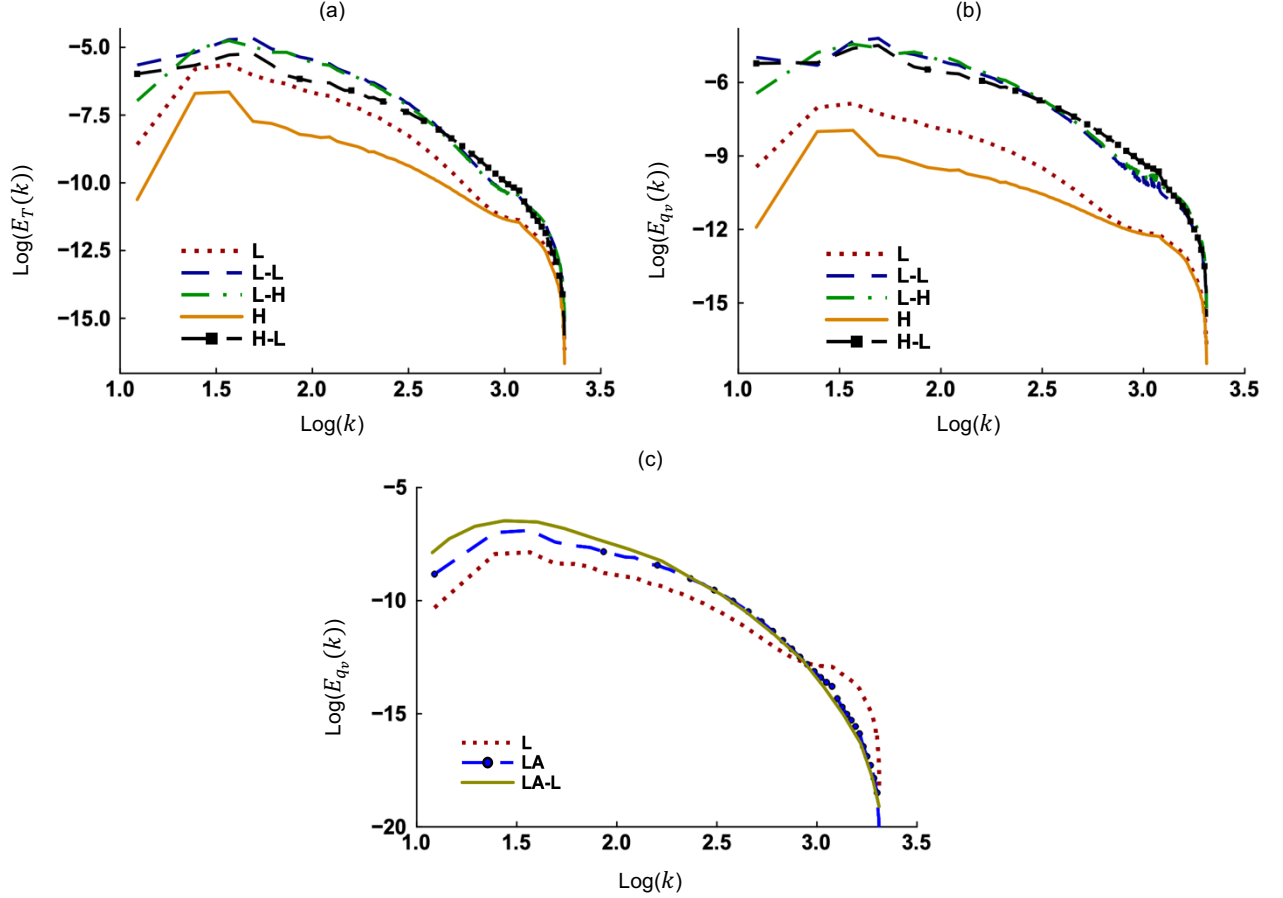


FIG. 3 Energy spectra of the (a) temperature and (b) vapor mixing ratio for Cases L, L-L, L-H, H, and H-L at 20 s in Scenario I. Energy spectra of the (c) vapor mixing ratio for Cases L, LA, and LA-L at 25 s in Scenario I. Note that the spectra of environmental supersaturation resemble those of vapor mixing ratio and hence are not shown. The wavenumbers (k) in the abscissa are multiples of k_{min} and are in the radial coordinate direction.

To examine the increase of fluctuations at small scales with presence of C_d (Fig. 3(c)), we plot the spectra of particle radius ($E_r(k)$) in Fig. 4(a). The particle field (r) gives feedback to the thermodynamic fields through the C_d . We notice that radius fluctuations are relatively significant at small scales but negligible at larger scales (Fig. 4(a)). Therefore, the $E_r(k)$ exhibits inverse energy cascade in our model. This also indicates that small-scale events such as diffusivity are modified by the presence of C_d and F_θ . To verify that, we solve the steady-state water vapor transport equation written as:

$$(\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_{v,eff} \nabla^2 q_v + F_{q_v}. \quad (21)$$

where $\mu_{v,eff}$ is the effective diffusivity for water vapor mixing ratio. Figure 4(b) shows that the $\mu_{v,eff}$ varies in space and deviates from a constant value of $2.2e-5$ (Table III). Thomas et al.¹⁹ suggested that diffusivity is differential for thermodynamic variables, although their LES based cloud model did not explicitly resolve small-scale interactions. Interestingly, the $\mu_{v,eff}$ is negative at many points (Fig. 4(b)) which implies the transfer of energy from small to large scales.³³ The spectra of $\mu_{v,eff}$ (Fig. 4(c)) confirm the backscatter. Hence, the energy spectra of forced and unforced thermodynamic fields do not converge at small scales (Fig. 3(b)) due to negative diffusivity and backscatter in our model. In real atmosphere, the modification of convective motion by condensational heating and evaporative cooling is an example of inverse (microscale to mesoscale) energy cascade.^{34,35}

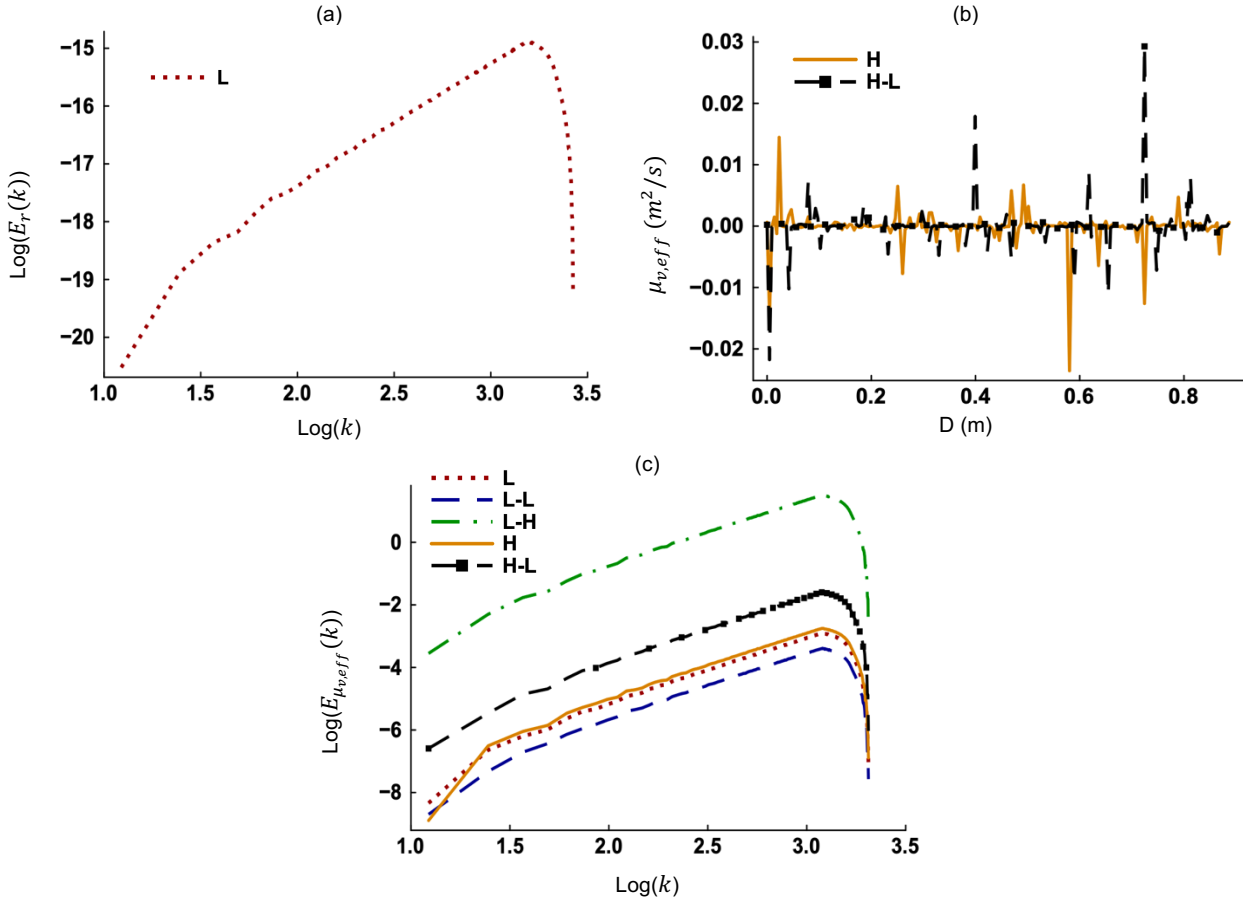


FIG. 4 (a) Energy spectra of the particle radii for Case L. (b) Spatial distribution of the effective vapor diffusivity for Cases L and L-H. (c) Energy spectra of the effective vapor diffusivity for all cases. These results correspond to Scenario I at 25 s. The wavenumbers (k) in the abscissa are multiples of k_{min} and are in the radial coordinate direction.

The correlation coefficient between the fluctuating vapor mixing ratio (q'_v) and fluctuating vertical velocity (w') is defined as:³⁶

$$R_{q'_v w'}(\mathbf{r}) = \frac{\langle q'_v(\mathbf{x}, t) w'(\mathbf{x} + \mathbf{r}, t) \rangle}{\sqrt{\langle (q'_v(\mathbf{x}, t))^2 \rangle \langle (w'(\mathbf{x}, t))^2 \rangle}}, \quad \mathbf{r} = \Delta x \mathbf{i} + \Delta y \mathbf{j} + \Delta z \mathbf{k}. \quad (22)$$

where $\mathbf{r}$ is the separation distance vector. Figure 5(a) shows that the q'_v and w' are not correlated for any of the cases. So, velocity fluctuations do not contribute to thermodynamic fluctuations in our model. Rather the latent heat release (or absorption) and thermodynamic forcing cause fluctuations in the thermodynamic fields. We see in Fig. 5(b) that the q'_v and $(F_{qv} - C_d)'$ are somewhat correlated at short distances for the thermodynamically forced cases (L-L, L-H, and H-L). At long distances, the coherent eddies break down into smaller structures and the correlation decays. The $R_{q'_v(F_{qv}-C_d)'}$ is weaker for the unforced cases (L and H) compared to the forced cases. Therefore, the C_d alone cannot sustain fluctuations if the F_{qv} is zero, consistent with Fig. 2.

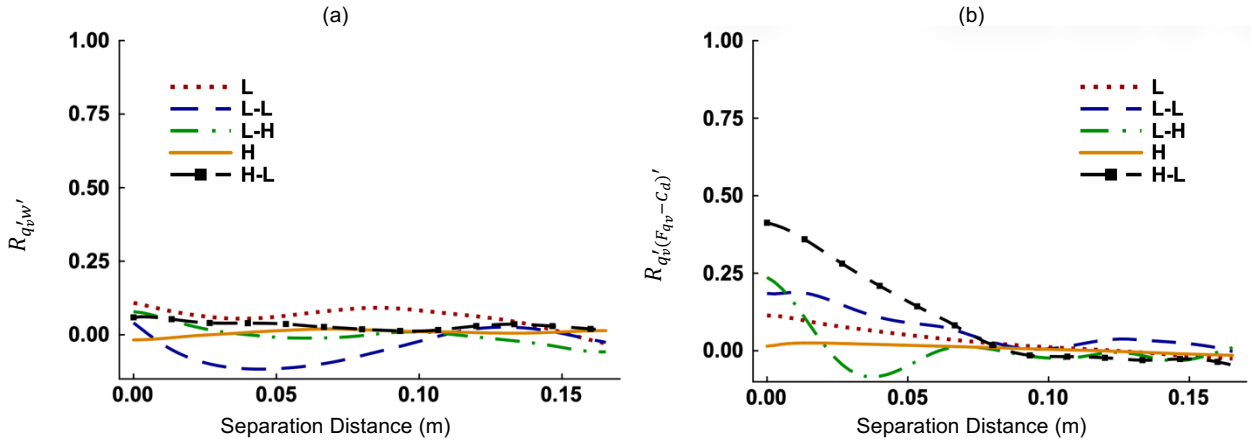


FIG. 5 The correlation coefficients for (a) fluctuating q_v and fluctuating w , and (b) fluctuating q_v and fluctuating $(F_{qv} - C_d)$. These results correspond to Scenario I at 25 s. Note that similar correlation profiles are observed for the temperature and environmental supersaturation.

Figure 6(a) shows the PDFs of T'/σ_T , which is the fluctuating temperature normalized by standard deviation, at 25 s in Scenario II. Figure 6(b) represents the same for vapor mixing ratio. It appears that none of the distributions in Figs. 6(a)-(b) are Gaussian. To measure the deviation from Gaussianity, we calculate the skewness (γ) and kurtosis (ω) of the probability distributions as³²

$$\gamma = \frac{\langle (\varphi')^3 \rangle}{\langle (\varphi')^2 \rangle^{3/2}}, \quad \omega = \frac{\langle (\varphi')^4 \rangle}{\langle (\varphi')^2 \rangle^{4/2}}, \quad \varphi = \frac{\theta'}{\sigma_\theta}. \quad (23)$$

The γ of temperature PDFs in Fig. 6(a) are 0.25, -0.55, -0.66, 0.22, and -0.19 for Cases L, L-L, L-H, H, and H-L, respectively, while the corresponding ω are 2.94, 4.14, 3.48, 2.09, and 3.26. The γ of vapor PDFs in Fig. 6(b) are -0.22, -0.86, -0.82, 0.21, and -0.62, respectively, while the corresponding ω are 2.87, 4.56, 4.05, 2.09, and 3.80. Skewness is 0 and kurtosis is 3 for a perfectly Gaussian distribution. We see that both temperature and vapor PDFs

are negatively skewed for the forced cases (L-L, L-H, and H-L) while they are positively skewed for the unforced cases (L and H). Negative skewness implies that most values are larger than the mean. Thus, thermodynamic forcing makes the magnitude of fluctuations larger than the average at most nodal points. We also find that the kurtosis of temperature and vapor PDFs are greater than 3 for the forced cases (L-L, L-H, and H-L) but less than 3 for the unforced cases (L and H). Hence, the PDFs are leptokurtic ($\omega > 3$) with a sharp peak (Figs. 6(a)-(b)) for Cases L-L, L-H, and H-L while they are platykurtic³⁷ ($\omega < 3$) with a rounded peak for Cases L and H. Previous studies of cloud turbulence reported that the PDFs of thermodynamic fluctuations can be non-Gaussian^{38,39} as this study shows. The magnitude of maximum T' in Figs. 6(a)-(b) is $\sim 4\sigma_T$ for Case L, $\sim 2\sigma_T$ for Case H, and $\sim 5\sigma_T$ for Cases L-L, L-H, and H-L. Again, the magnitude of maximum q'_v is $\sim 3\sigma_{q_v}$ for Case L, $\sim 2\sigma_{q_v}$ for Case H, $\sim 5\sigma_{q_v}$ for Cases L-L and H-L, and $\sim 6\sigma_{q_v}$ for Cases L-H. Similar ranges were reported by Anderson et al.⁴⁰ in their experimental investigation of cloud processes. They quantified that the maximum T' is $\sim 4\sigma_T$ and maximum q'_v is $\sim 6\sigma_{q_v}$ in the updraft region of cloud.

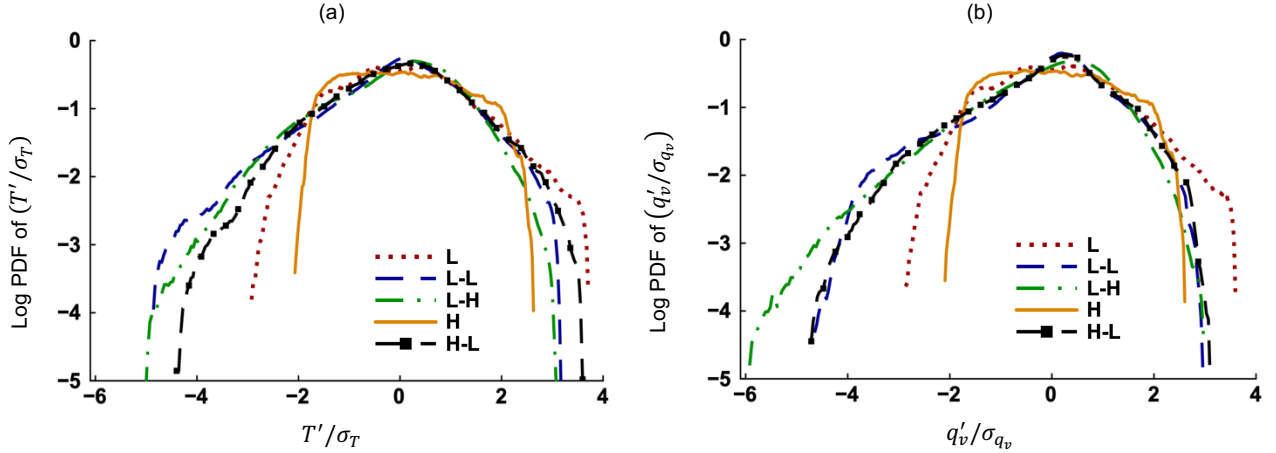
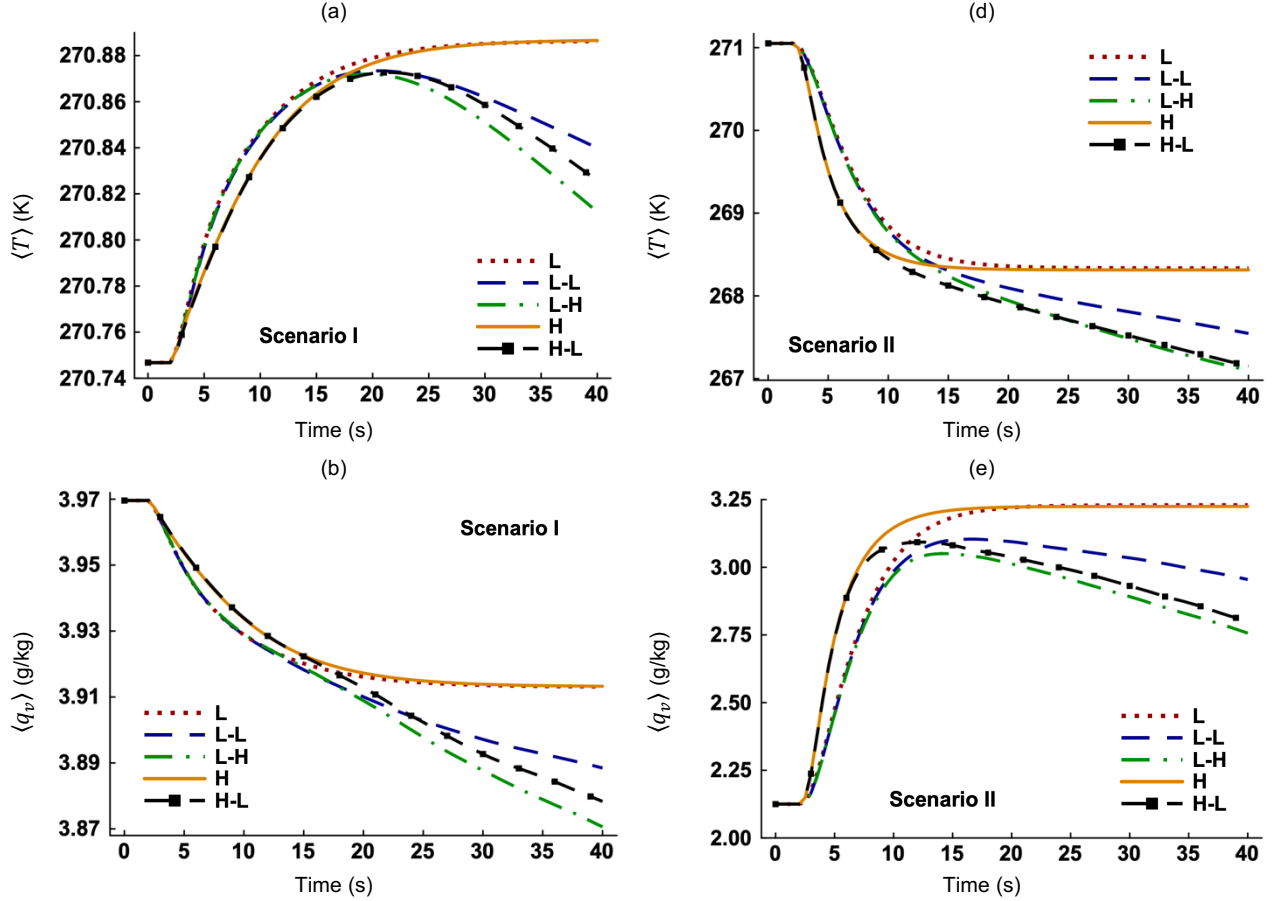


FIG. 6 Log PDFs of the (a) fluctuating temperature normalized by standard deviation (T'/σ_T), and (b) fluctuating vapor mixing ratio normalized by standard deviation (q'_v/σ_{q_v}) in Scenario II at 25 s. Note that the PDFs of (S'_e/σ_{S_e}) resemble that of (q'_v/σ_{q_v}) and hence are not shown.

Thermodynamic and momentum forcings influence the evolution of mean thermodynamic fields in our model (Figs. 7(a)-(f)). The mean temperature ($\langle T \rangle$) increases and decreases during condensation (Fig. 7(a)) and evaporation (Fig. 7(d)), respectively, and reaches an asymptote for Cases L and H. The mean water vapor mixing ratio ($\langle q_v \rangle$) decreases in Scenario I (Fig. 7(b)) but increases in Scenario II (Fig. 7(e)) for Cases L and H. This is because water vapor converts into liquid water during condensation and the converse during evaporation. The profiles of mean supersaturation ($\langle S_e \rangle$) in Figs. 7(c) and 7(f) resemble those of the $\langle q_v \rangle$ since the $\langle S_e \rangle$ is linearly proportional to the $\langle q_v \rangle$. In absence of thermodynamic forcing, the system reaches saturation ($\langle S_e \rangle \sim 0$) in both scenarios although their

initial conditions differ. The difference between thermodynamically forced and unforced cases becomes prominent after 20 s in Scenario I and after 15 s in Scenario II. The increase of $\langle T \rangle$ in Scenario I (Fig. 7(a)) is followed by a decrease which happens faster if the level of thermodynamic forcing is high (Case L-H). The same pattern is observed for the $\langle q_v \rangle$ in Scenario II (Fig. 7(e)). The $\langle T \rangle$ continues to decrease for the forced cases in Scenario II (Fig. 7(d)) and the same is true for the $\langle q_v \rangle$ in Scenario I (Fig. 7(b)). We find that the mean profiles are impacted by momentum forcing until saturation is reached. The $\langle S_e \rangle$ changes faster in Scenario II for a high level of momentum forcing (Case H) compared to Case L (Fig. 7(f)). The converse is true in Scenario I (Fig. 7(c)). Strong turbulence (Case H) enhances mixing between the cloud and dry air in Scenario II, leading to a higher evaporation rate and quicker increase of water vapor (Fig. 7(e)). But strong mixing reduces supersaturation in the cloudy region and suppresses condensation. Thus, water vapor decreases slowly for Case H relative to Case L in Scenario I (Fig. 7(b)).



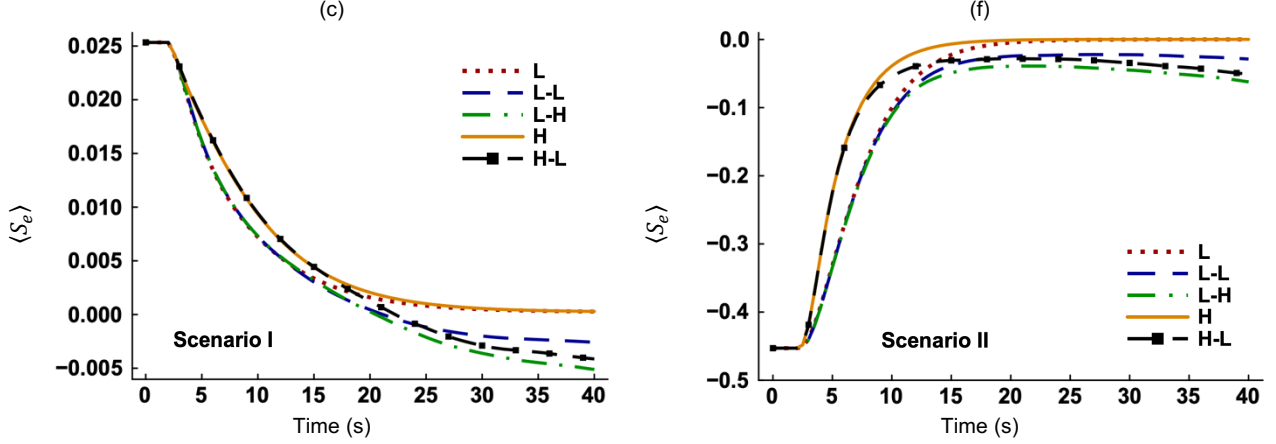


FIG. 7 Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), respectively in Scenario I. Time evolution of the (d) mean temperature ($\langle T \rangle$), (e) mean vapor mixing ratio ($\langle q_v \rangle$), and (f) mean environmental supersaturation ($\langle S_e \rangle$), respectively in Scenario II.

B. The impact of thermodynamic forcing on microphysics

1. Condensation and activation (Scenario I)

Aerosol particles grow and activate into cloud droplets in Scenario I. Figure 8(a) shows the temporal evolution of the mean radii ($\langle r \rangle$) of all particles (aerosol and cloud). The $\langle r \rangle$ increases exponentially at early times for all cases, corresponding to positive supersaturation (Fig. 7(c)). Particle growth levels off for the unforced cases (L and H) by 25 s while it becomes negative for the forced cases (L-L, L-H, and H-L) after 22 s. Hence, condensation is suppressed in our model if thermodynamic forcing is present. We see in Fig. 8(b) that the number of aerosol particles drops to nearly zero for all cases as they grow beyond the critical radius of $1.26 \mu\text{m}$ and activate into cloud droplets. According to Fig. 8(c), the relative dispersion of particle radii ($\sigma_r / \langle r \rangle$) reaches a peak around 5 s for low turbulence cases (L, L-L, and L-H) and around 3 s for high turbulence cases (H and H-L). So, flow turbulence (level of momentum forcing) affects the peak of $\sigma_r / \langle r \rangle$. The $\sigma_r / \langle r \rangle$ decreases from the peak value and levels off for Cases L and H as condensational growth slows down. For Cases L-L, L-H, and H-L, the decrease of $\sigma_r / \langle r \rangle$ is followed by another increase. Although $S_e \geq 0$ in Scenario I, thermodynamic forcing leads to $S_e < 0$ and thus evaporation at some nodal points. This increases the standard deviation of droplet radii (σ_r). Figure 8(d) shows the time evolution of $\langle r' S'_e \rangle$ which correlates r' with S'_e . The $\langle r' S'_e \rangle$ increases at early times and then decays to zero if thermodynamic forcing is absent while it becomes negative for the forced cases. The decay of this correlation signifies turbulent homogenization in our model. Again, the peak of $\langle r' S'_e \rangle$ is higher if turbulent mixing (momentum forcing) is stronger (Cases H and H-L).

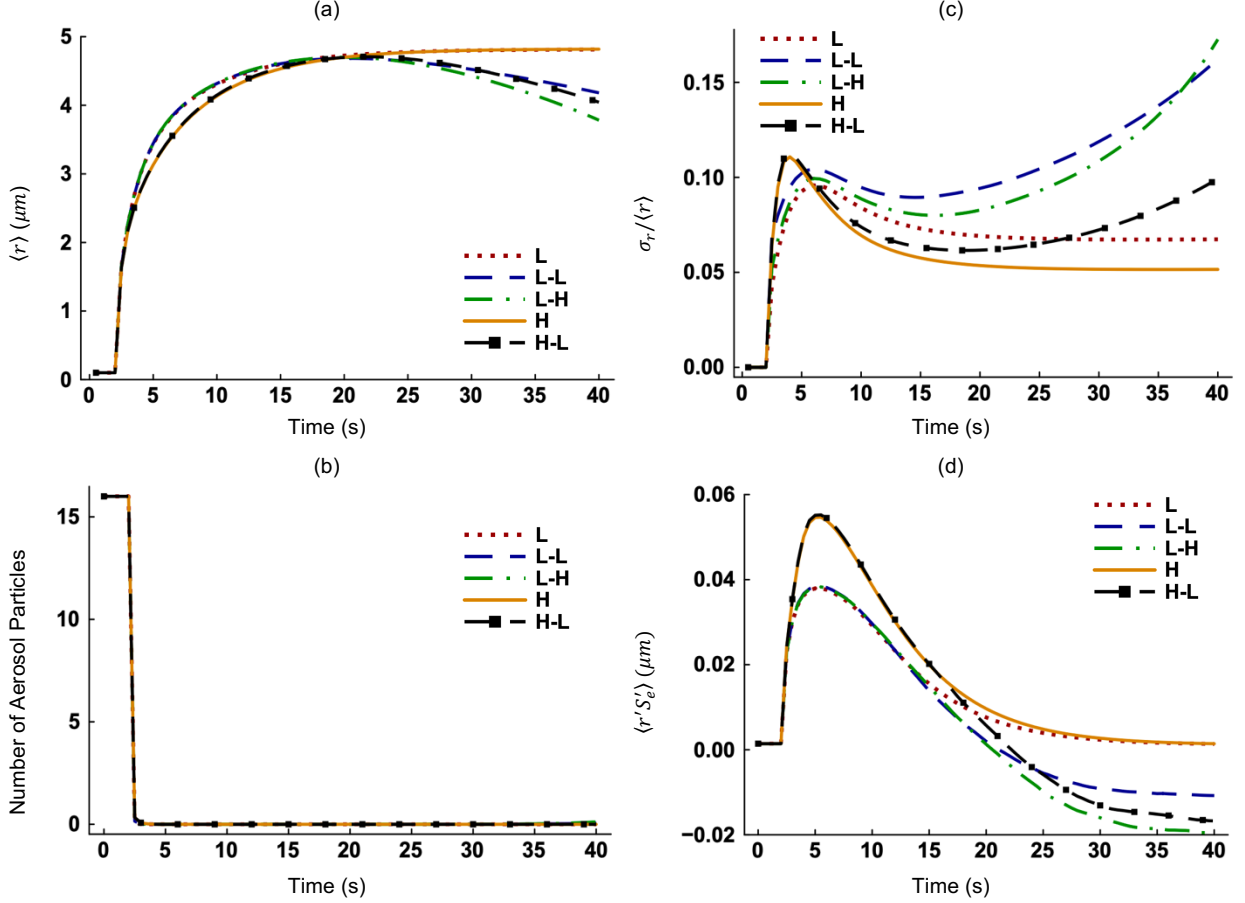


FIG. 8 Time evolution of the (a) mean radius ($\langle r \rangle$) of all particles, (b) number of aerosol particles, (c) relative dispersion of particle radii ($\sigma_r / \langle r \rangle$), and (d) one-point correlation ($\langle r' S_e' \rangle$) in Scenario I. No change is observed in first two seconds as particles are allowed to grow after 2 s.

The PDFs of particle radii (f_r) are shown in Figs. 9(a)-(d) at selected times in Scenario I. Only dry aerosols, each having a radius of $0.1 \mu m$, exist at 0 s. Their radius PDFs are represented by the dotted vertical lines in Figs. 9(a)-(d). By 10 s, all of them activate into cloud droplets for Case L (Fig. 9(a)) and the radius PDF appears to be negatively skewed. The f_r shifts to the right as cloud droplets grow bigger by 25 s. No significant growth takes place between 25 s and 35 s and their f_r converge (Fig. 9(a)). Convergence of the f_r^{41} is also observed in Fig. 9(d) if flow turbulence increases but no thermodynamic forcing is applied (Case H). Although the mean radii are identical for Cases L and H at 35 s (Fig. 8(a)), the f_r is narrower for the latter (Fig. 9(d)). Inclusion of thermodynamic forcing (Cases L-H and L-L) increases the difference between maximum and minimum radii and broadens the f_r (Figs. 9(b)-(c)). The f_r produced by models are narrower than those observed in experiments.⁴² So, broadening of the f_r due to thermodynamic forcing improves the accuracy of numerical models. Interestingly, all particles are larger than the critical radius (r_c) at 10 s but some of them are smaller than the r_c at 25 s and 35 s for Cases L-L and L-H. That is, a small fraction of the

previously activated cloud droplets deactivate into aerosol particles at late times if thermodynamic forcing is applied.

For the forced cases, the spread of f_r does not change between 25 s and 35 s (Figs. 9(b)-(c)).

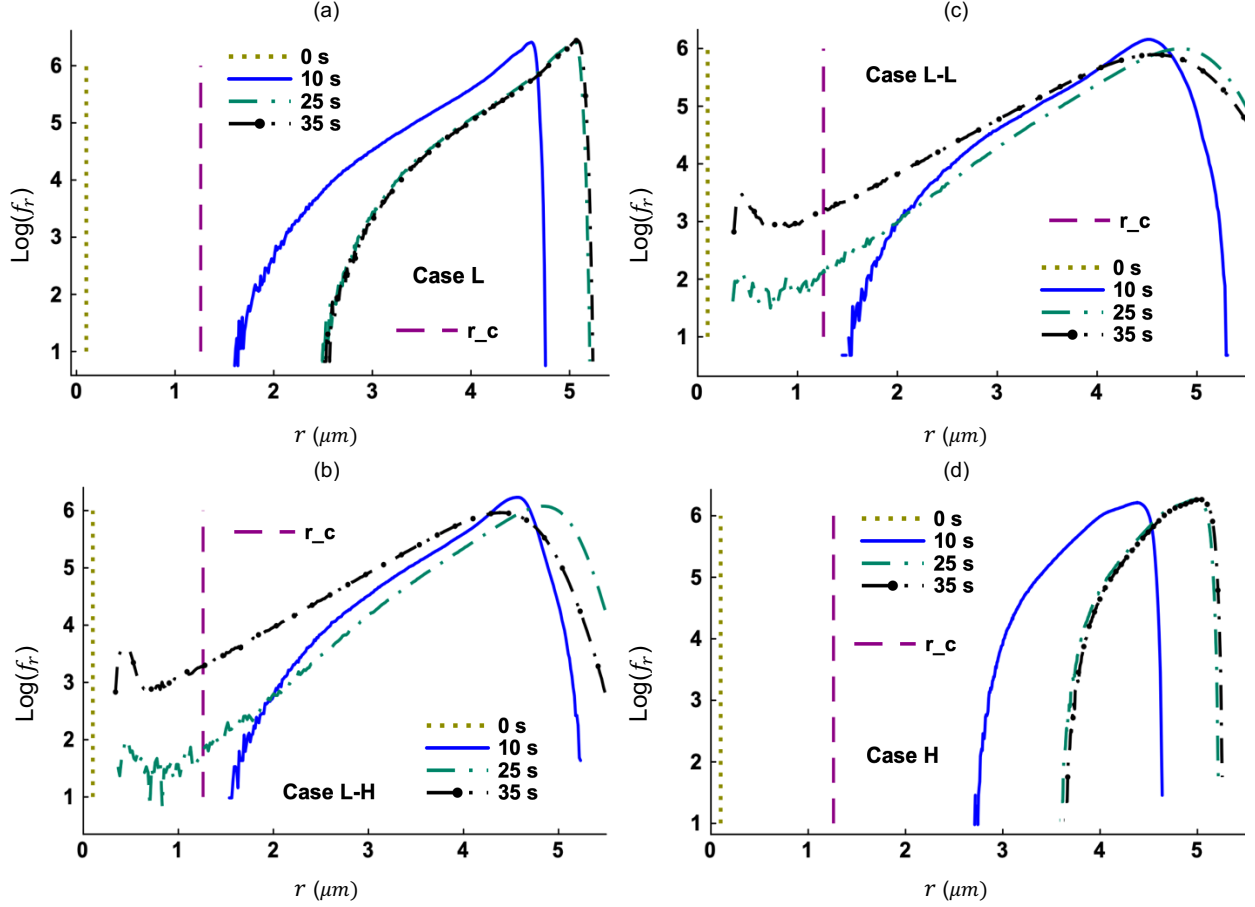


FIG. 9 Log PDFs of particle radii for (a) Case L, (b) Case L-H, (c) Case L-L, and (d) Case H at selected times in Scenario I. The critical radius (r_c) is marked by the dashed vertical line.

2. Evaporation and deactivation (Scenario II)

Cloud droplets have a monodisperse (15 μm) size distribution initially in Scenario II and experience evaporative decay with time. Figure 10(a) indicates that the mean radii ($\langle r \rangle$) decreases faster for the forced cases (L-L, L-H, and H-L) compared to the unforced cases (L and H). Hence, evaporation is stronger in our model if thermodynamic forcing is applied. A higher level of thermodynamic forcing (Case L-H) enhances evaporation (Fig. 10(a)) relative to low thermodynamic forcing (Case L-L). We find that the decay of $\langle r \rangle$ is quicker for Case H compared Case L and for Case H-L compared to Cases L-H and L-L. Thus, evaporation is enhanced if the turbulence intensity increases regardless of thermodynamic forcing. The number of cloud droplets drops over time (Fig. 10(b)) as evaporation and deactivation continues. The drop is larger for Case L-H compared to Case H-L, meaning that more cloud droplets deactivate into

aerosol particles for the former. The same is observed between Cases L and H. So, a higher level of flow turbulence reduces deactivation rate in our model. Note that individual particle statistics is considered to calculate deactivation rate (Fig. 10(b)) while domain-averaged data is used to calculate evaporation rate (Fig. 10(a)). Deactivation is strongly suppressed for Case H (Fig. 10(b)) as most cloud droplets do not evaporate enough to fall below the critical barrier (r_c). We also notice that thermodynamic forcing increases the rate of deactivation for any turbulence level. Figure 10(c) shows the time evolution of the $\sigma_r/\langle r \rangle$ in Scenario II. The $\sigma_r/\langle r \rangle$ increases for the thermodynamically forced cases, for which cloud droplets exist in various sizes due to partial evaporation. The relative dispersion becomes steady for the unforced cases when saturation is reached and evaporation stops. The r' and S'_e are negatively correlated at initial time for all cases in Scenario II (Fig. 10(d)). The correlation $\langle r'S'_e \rangle$ weakens over time due to mixing. At long time, the $\langle r'S'_e \rangle$ becomes zero for Cases L and H but remains slightly negative for Cases L-L, L-H, and H-L.

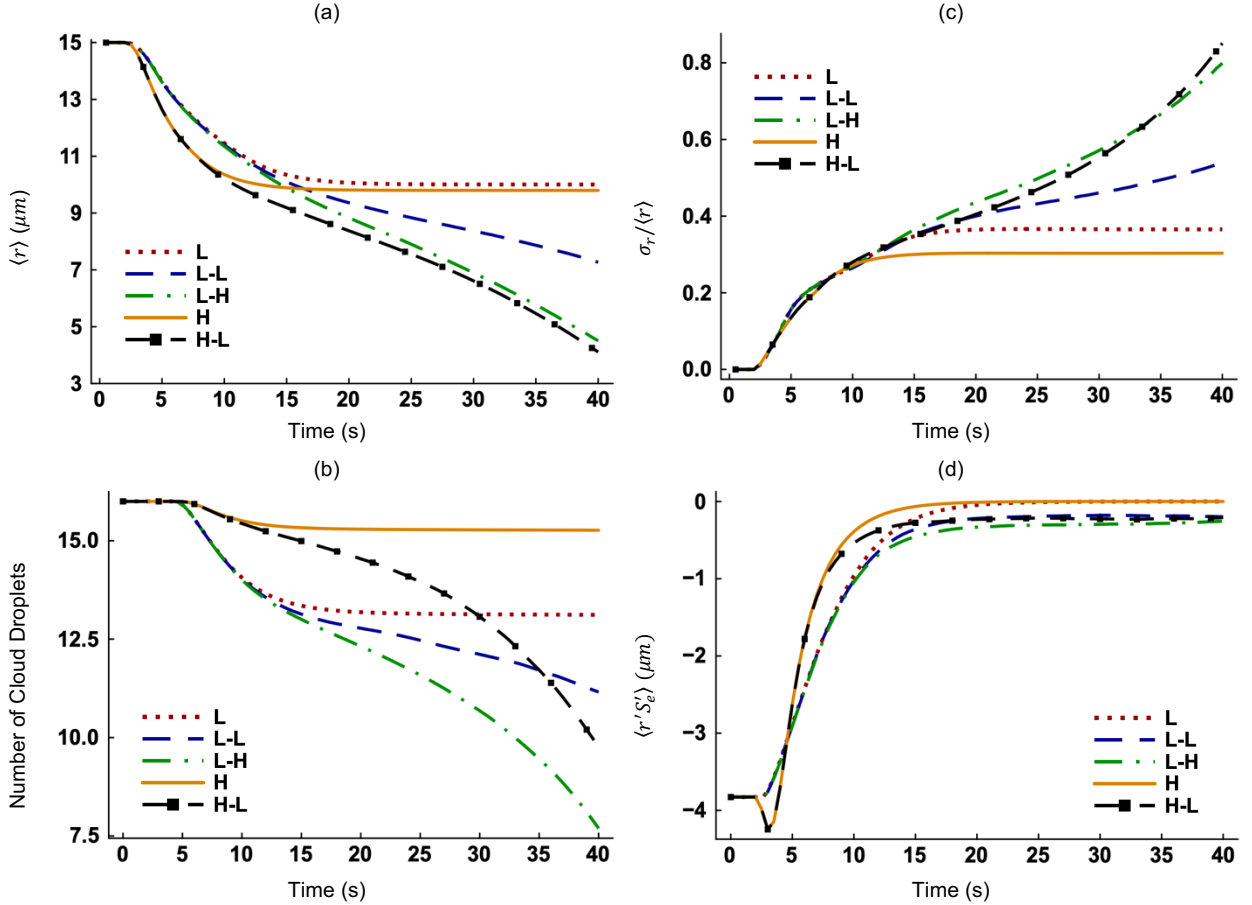


FIG. 10 Time evolution of the (a) mean radius ($\langle r \rangle$) of all particles, (b) number of aerosol particles, (c) relative dispersion of particle radii ($\sigma_r/\langle r \rangle$), and (d) one-point correlation ($\langle r'S'_e \rangle$) in Scenario II. No change is observed in first two seconds as particles are allowed to grow after 2 s.

The radius PDFs in Scenario II appear to be bimodal for all cases (Figs. 11(a)-(d)), with one peak to the left of the r_c line and another to the right. The existence of bimodality⁴³ indicates inhomogeneous evaporation. A group of cloud droplets experiences strong evaporation, becomes much smaller, and deactivates into aerosol particles. These particles correspond to the left peak while the right peak is formed by the rest of the droplet population. Manton⁴⁴ suggested that flow turbulence causes cloud droplets to grow (or shrink) differently and it creates a bimodal particle size distribution as shown by our model. Some droplets evaporate very little by 35 s and their sizes are very close to the initial size of $15 \mu\text{m}$ for Case L (Fig. 11(a)). If thermodynamic forcing is applied (Case L-L), a small number of cloud droplets grows instead of shrinking (Fig. 11(c)). This makes the f_r broader compared to the unforced case (L). The f_r narrows for Case H (Fig. 11(b)) as the turbulence intensity increases and more cloud droplets undergo complete evaporation. If thermodynamic forcing is added to a high turbulence condition (Case H-L), complete evaporation is promoted further and the maximum cloud droplet radii decrease at all three times (Fig. 11(d)).

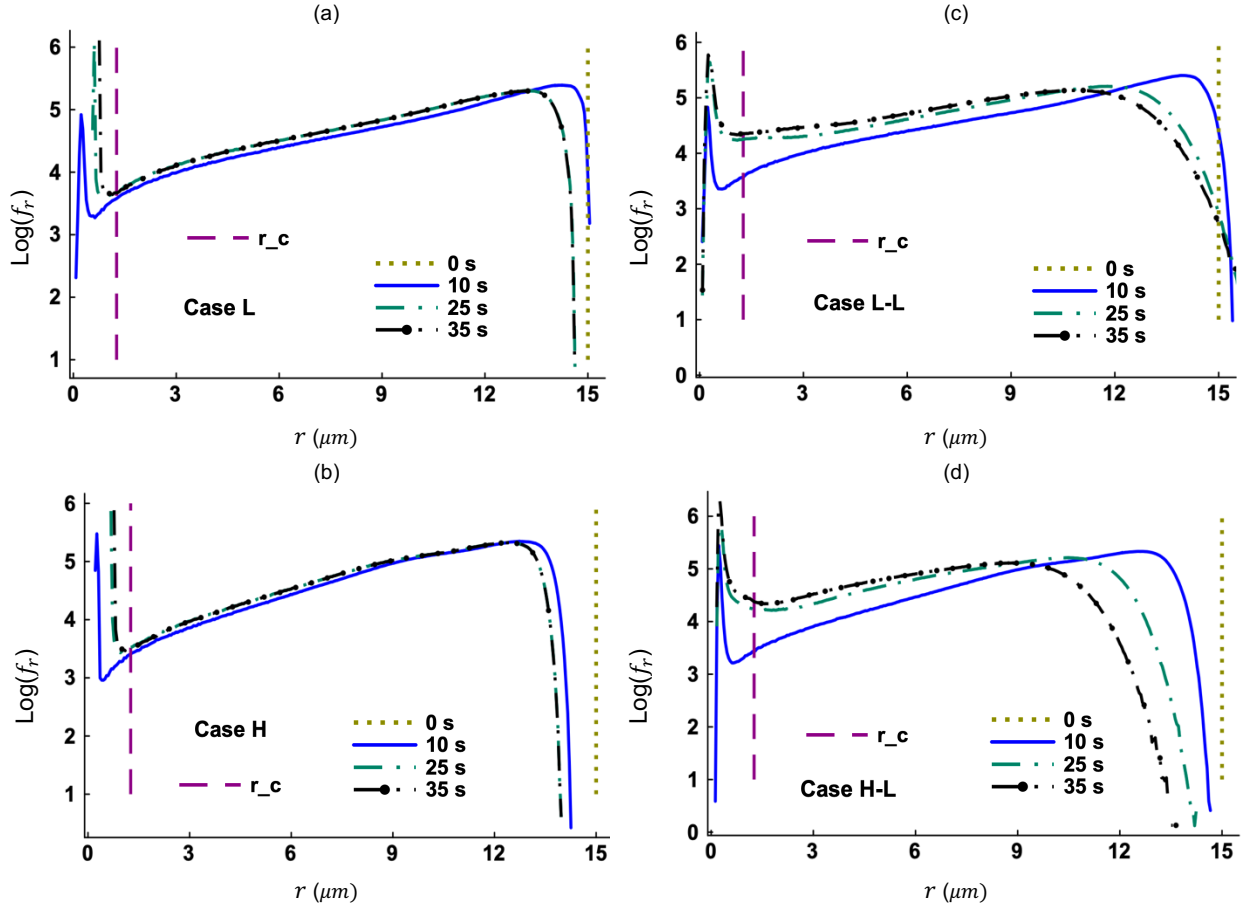


FIG. 11 Log PDFs of particle radii for (a) Case L, (b) Case H, (c) Case L-L, and (d) Case H-L at selected times in Scenario II. The critical radius (r_c) is marked by the dashed vertical line.

V. CONCLUDING REMARKS

A three-dimensional particle-based direct numerical simulation (DNS) model is employed with inclusion of low-wavenumber forcing in the momentum and thermodynamic fields.⁴⁵ Entrainment and turbulent mixing between the cloud and environment is simulated. The key findings from our simulations are itemized below.

- The fluctuating thermodynamic fields become statistical stationarity (Figs. 2(a)-(d)) if external thermodynamic forcing is applied but decay otherwise. The decay is caused by turbulent mixing and homogenization. A higher level of momentum forcing accelerates the decay.
- Thermodynamic forcing reduces the mean temperature and mean vapor mixing ratio during condensation and evaporation (Figs. 7(a)-(b) and 7(d)-(e)). Strong turbulence or high level of momentum forcing drives the system to saturation faster in both scenarios (Figs. 7(c) and 7(f)).
- Although thermodynamic forcing is applied at large scales only, it affects the energy spectra of thermodynamic fields at all scales (Figs. 3(a)-(b)) by interacting with the phase change term. The back transfer of energy from small to large scales is evident in the spectra of particle radius (Fig. 4(a)) and water vapor diffusivity (Fig. 4(c)).
- Velocity fluctuations do not have significant impact on thermodynamic fluctuations in our model (Fig. 5(a)). External thermodynamic forcing is the primary source of thermodynamic fluctuations (Fig. 5(b)).
- The PDFs of fluctuating temperature and vapor fields are negatively skewed for the thermodynamically forced cases but positively skewed for the unforced cases (Figs. 6(a)-(b)). The kurtosis of temperature and vapor PDFs are greater than 3 for the forced cases but less than 3 for the unforced cases.
- Thermodynamic forcing suppresses condensation (Fig. 8(a)) by creating locally subsaturated conditions in our model although the overall environment is supersaturated. It increases the relative dispersion of particle radii for all turbulence levels (Fig. 8(b)).
- The PDFs of particle radii become narrower during condensation if the turbulence intensity increases and no thermodynamic forcing is applied (Fig. 9(d)). With thermodynamic forcing, a small fraction of the previously activated cloud droplets deactivates into aerosol particles, and the radius PDFs broaden (Figs. 9(b)-(c)).
- Thermodynamic forcing enhances evaporation (Fig. 10(a)) and deactivation (Fig. 10(b)) in our model. A high level of momentum forcing enhances mean evaporation (Fig. 10(a)) but suppresses deactivation of individual cloud droplets (Fig. 10(b)).

- The radius PDFs are found to be bimodal during evaporation (Figs. 11(a)-(d)). They are broader in response to thermodynamic forcing if the turbulence intensity is low (Fig. 11(c)). More cloud droplets evaporate completely if the turbulence level increases, leading to a decrease of the maximum droplet radii (Figs. 11(b) and 11(d)).

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DATA AVAILABILITY

The data supporting the findings of this work are available from the corresponding author upon reasonable request.

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